Olivia Kaitlyn Roberts, MS – Curriculum Vitae

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EDUCATION

University of Michigan, Ann Arbor, MI — Aug 2024 – May 2026
M.S. Computational Epidemiology and Systems Modeling
Certificate in Complex Systems

University of South Dakota, Vermillion, SD — Aug 2021 – May 2024
B.S. Mathematics
Minors in Music and Interdisciplinary Sciences

RESEARCH EXPERIENCE

Research Assistant II, Center for the Assessment of Tobacco Regulations (CAsToR) — May 2025 – Present

- Conducts research on tobacco use trends with Dr. Andrew Brouwer on the Data Analysis and Modeling team
- Applies multistate transition models using complex survey weights in R to analyze longitudinal trends in tobacco use among adults and youth

UDiscover Summer Research Scholar, University of South Dakota — May 2023 – Aug 2023

- Conducted pure mathematics research under Dr. Gabriel Picioroaga on generalizing music theory using algebra
- Developed mathematical proofs and Maple programs to construct new musical systems

Behavioral Health Epidemiology Intern, South Dakota Department of Social Services — May 2022 – Aug 2022

- Analyzed state-level data on substance use and mental health using R, Excel, and Power BI
- Contributed to behavioral health data and outcomes reports for state government and public use

TECHNICAL SKILLS

- Programming: R, Python, Matlab, Stata
- Tools: LaTeX, Power BI, Excel, SEER Stat, SQL
- Markov multistate transition models
- Network models: contact networks, outbreak simulation
- Infectious and chronic disease models: deterministic and stochastic models

PUBLICATIONS

- Picioroaga G, **Roberts O**. Musical systems with $\mathbb{Z}_n$ – Cayley graphs. *MusMat Brazilian Journal of Music and Mathematics.* 2024;8(1). doi:10.46926/musmat.2024v8.1-23.
- Brouwer AF, **Roberts OK**, Jeon J, Jimenez-Mendoza E, Land SR, Freedman ND, Torres-Alvarez R, Mistry R, Levy DT, Meza R. Transitions in cigarette and ENDS use in the PATH Study: a multistate transition model analysis of adults in 2021–2022 compared to previous years. *Manuscript submitted*
- **Roberts OK**, Jeon J, Jimenez-Mendoza E, Land SR, Freedman ND, Torres-Alvarez R, Mistry R, Levy DT, Meza R, Brouwer AF. Transitions in cigarette, ENDS, and dual use among youth in the PATH Study from 2015–2023: a multistate transition modeling analysis. *Manuscript in preparation*
- **Roberts OK**, Bover Manderski MT, Gonsalves NJ, Jeon J, Meza R, Delnevo CD, Brouwer AF. Trends in transitions in cigarette and electronic nicotine vapor product use in the Rutgers Omnibus Study, 2022–2025. *Manuscript in preparation*

PRESENTATIONS

"Trends in transitions in cigarette and electronic nicotine vapor product use in the Rutgers Omnibus study, 2022–2025"

- Poster presentation, University of Michigan Epidemiology Poster Session, April 2026

"Transitions in cigarette, ENDS, and dual use among youth in the PATH Study from 2015–2023: a multistate transition modeling analysis"

- Invited seminar talk (1 hour), University of South Dakota Department of Mathematics, March 2026
- Oral presentation, Society for Research on Nicotine and Tobacco (SRNT) Annual Meeting, March 2026

"Musical Systems with $\mathbb{Z}_n$ – Cayley Graphs"

- Oral presentation, Joint Mathematics Meetings (JMM) Pi Mu Epsilon Session, Jan 2024
- Oral presentation, Nebraska Conference for Undergraduate Women in Mathematics (NCUWM), Jan 2024
- Oral presentation, National Conference on Undergraduate Research (NCUR), April 2023 and April 2024

HONORS & AWARDS

Research Awards and Grants

- Pi Mu Epsilon Outstanding Student Speaker Award, Joint Mathematics Meetings, Jan 2024
- Undergraduate Research Excellence Award, University of South Dakota, Sept 2023
- UDiscover Summer Research Grant, University of South Dakota, May 2023
- Carlson Research Grant, University of South Dakota, April 2023

Scholarships

- Epidemiology Student Scholarship, University of Michigan, Feb 2024 (75% of tuition)
- South Dakota NASA Space Grant Consortium Educational Stipend, Oct 2023
- Cleland Cook Mathematics Scholarship, University of South Dakota, April 2022 and April 2023

TEACHING EXPERIENCE

Yoga Teacher, University of Michigan Recreation — Aug 2024 – Present

- Leads restorative and vinyasa yoga classes to 20+ students multiple times a week
- Developed a 6-week Elevate Yoga workshop series to deepen students' yoga practice

Instructor Aide, University of Michigan Department of Mathematics — July 2024 and July 2025

- Co-instructed three summer camp sessions for 15-20 high school students for the Michigan Math and Science Scholars program
- Assisted with curriculum development and delivered lectures on group theory, set theory, and music theory

Instructor Assistant, University of Michigan School of Nursing — Jan 2025 – May 2025

- Supported 100 students in NURS 218 (Applied Statistics)
- Held office hours, graded assignments, assisted students with Stata programming, and led review sessions

Math Tutor, University of South Dakota — Jan 2022 – May 2024

- Tutored undergraduate students in algebra, statistics, and calculus

Yoga Teacher, University of South Dakota Wellness Center — Aug 2021 – May 2024

- Lead sculpt and vinyasa yoga classes for 20+ students multiple times per week
- Organized the first and second annual USD Yoga Festivals and lead multiple yoga workshops

OTHER

Event Volunteer, VegMichigan June 2024 – Present

Admissions Ambassador, University of Michigan School of Public Health Aug 2024 – May 2026

200-Hour Yoga Teaching Certification, Tiny Buddha Yoga, Dec 2024

200-Hour Yoga Teaching Certification, Better Together Yoga, Sept 2021